QuickSyn session report

session

QuickSyn Session Reduction and Veusz Export 0.0.9

Session 2026-10-12 18:02:04 UTC (UTC+0000) to 2026-10-12 18:12:04 UTC (UTC+0000)

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1 Instrument

Field	Value
`*IDN?` reply	`Phase Matrix,FSL-0020,0000017f,1530200407,0008`
Model	FSL-0020
Option word	`0000017f` (options 1, 2, 3, 4, 5, 6, 7, 9)
Serial number	`1530200407`
Firmware version	`0008`
Datasheet coverage	0.65 to 20 GHz
Transport	COM5 at 115200 baud

`*IDN?` fields: manufacturer, model, option word, serial, firmware. [1] does not define the order; it was confirmed against NI's soft front panels on an FSL-0020 (option bit n = option $n+1$).

2 Acquisition timing

Quantity	Value	Unit
Samples	1201	
Loaded from	chunks	
Duration	600	s
Requested sample rate	2	Hz
Achieved sample rate	2	Hz
Mean sample interval	500	ms
Interval standard deviation (jitter)	14.8413	ms
Interval minimum	312	ms
Interval maximum	688	ms
Jitter relative to the period	29682.7	ppm
Device I/O latency, mean	11.7086	ms
Device I/O latency, 95th percentile	16	ms
Device I/O latency, maximum	188	ms

Timing of the telemetry stream.

The sample period is the reciprocal of the requested rate f_s,

$$\tau_0 = \frac{1}{f_s}, \tag{1}$$

and the achievable rate is bounded by the serial round trips each sample costs. With N_q queries per sample and a mean round-trip time t_q,

$$f_{s,\max} = \frac{1}{N_q\, t_q}. \tag{2}$$

The jitter is the standard deviation of the measured intervals,

$$\sigma_{\tau_0} = \operatorname{std}\left(t_i - t_{i-1}\right), \tag{3}$$

taken from the monotonic clock of the logging computer.

3 Generated signal

Quantity	Value	Unit
Distinct frequencies observed	1	
Frequency minimum	1.000000000	GHz
Frequency maximum	1.000000000	GHz
Power minimum	-10.0	dBm
Power maximum	+5.0	dBm
RF output on, fraction of samples	0.127	
Samples with RF unlocked	0	
Samples with reference unlocked	0	
Samples with voltage error	0	

Signal state reported by the instrument.

Status bits are decoded from `STAT?` with the bit map of [1, Table 5]; the lock bits are set when *unlocked*.

4 Thermal

Quantity	Value	Unit
Internal temperature minimum	26.8	°C
Internal temperature maximum	57.5	°C
Rise over the session	+2.1	°C
Margin to the +75 °C limit	17.5	°C

Internal temperature (DIAG:MEAS? 21).

The +75 °C internal limit is the caution in [2].

5 Stability of the reported carrier

The fractional frequency error of sample i against the median commanded carrier f_c is

$$y_i = \frac{f_i - f_c}{f_c}, \tag{4}$$

and the overlapping Allan deviation at averaging time $\tau = m\,\tau_0$ follows IEEE Std 1139-2008 [4]:

$$\sigma_y^2(\tau) = \frac{1}{2\,(N - 2m + 1)\,\tau^2} \sum_{k=1}^{N-2m+1} \left(x_{k+2m} - 2x_{k+m} + x_k\right)^2, \tag{5}$$

with the phase residuals

$$x_k = \tau_0 \sum_{i<k} y_i. \tag{6}$$

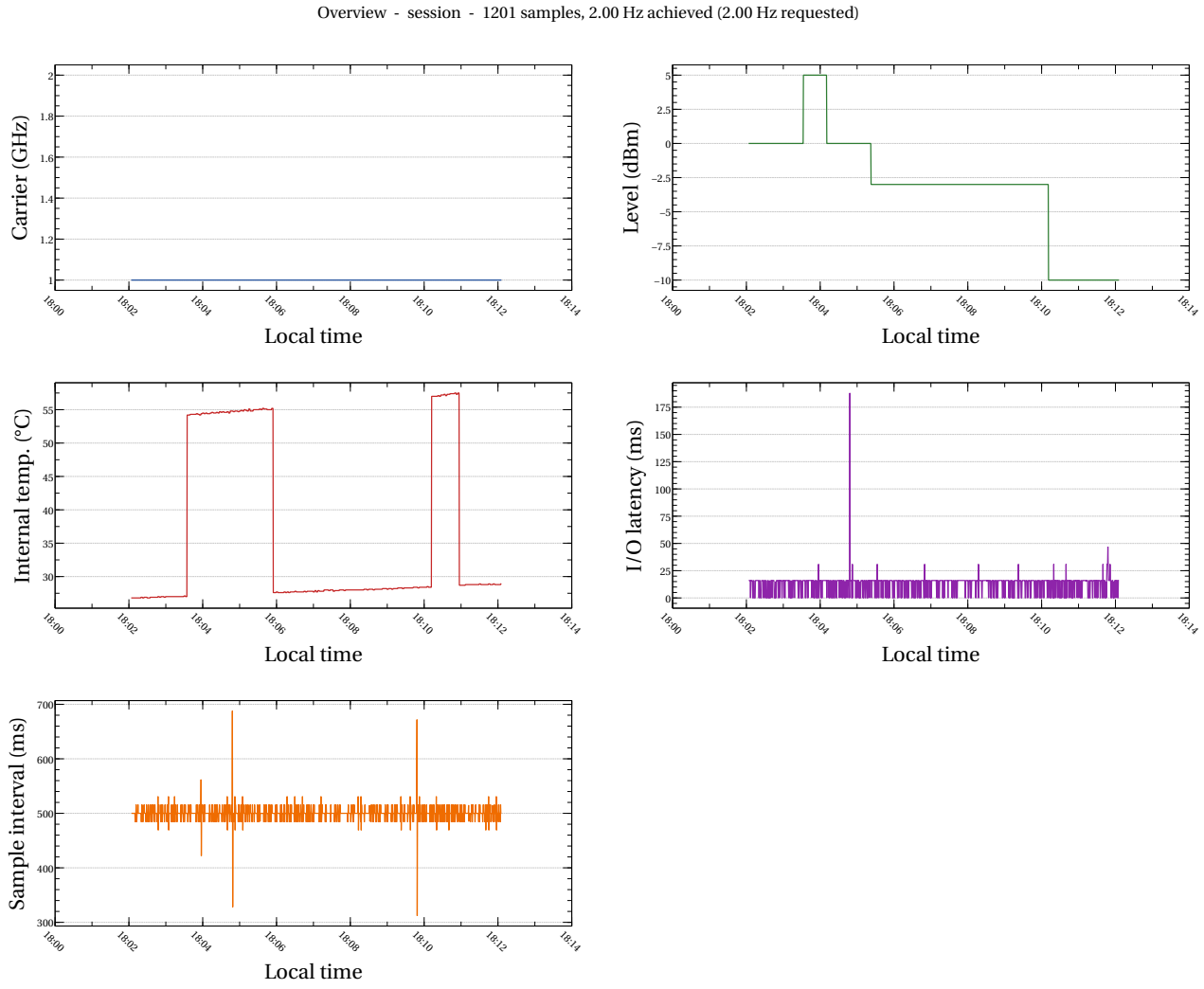
This characterises command fidelity and the logging time base, not the phase noise of the carrier: `FREQ?` returns the commanded setting, not a measurement of the output. Phase noise is specified separately in the datasheet [3] and cannot be inferred from this log.

Averaging time τ (s)	**ADEV $\sigma_y(\tau)$**
0.5	0.000e+00
1	0.000e+00
2	0.000e+00
4	0.000e+00
8	0.000e+00
16	0.000e+00
32	0.000e+00
64	0.000e+00
128	0.000e+00
256	0.000e+00

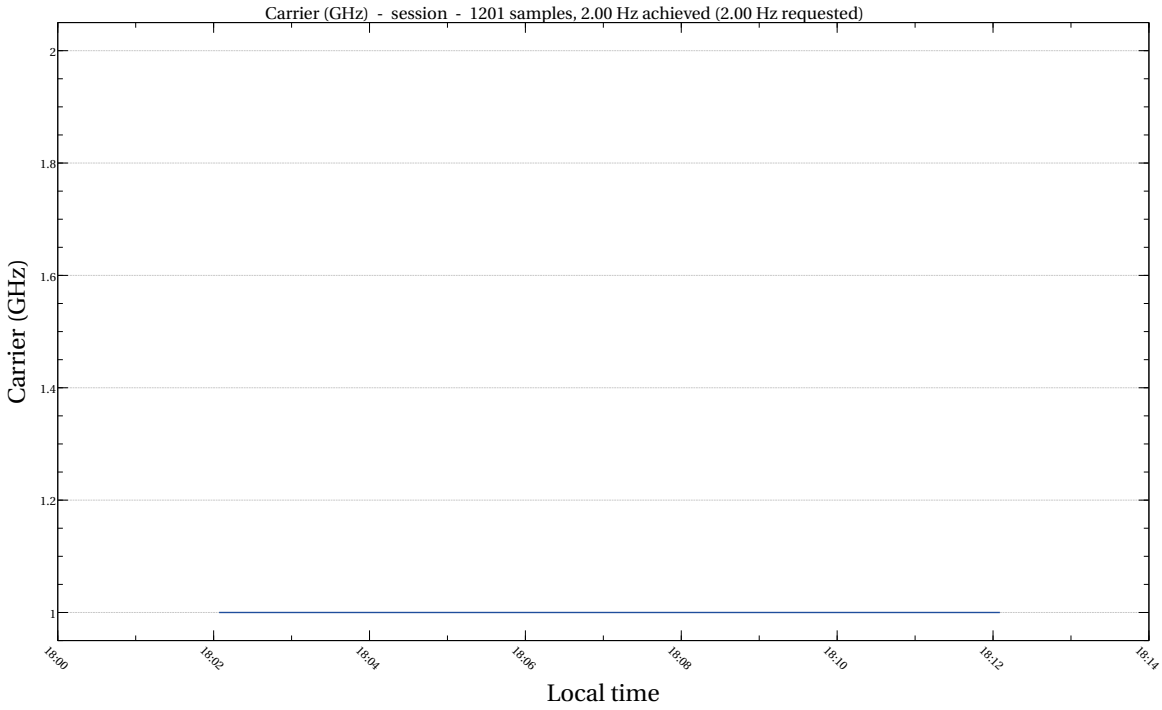
6 Plots

Every page of `quicksyn_session.pdf`, rendered by Veusz [5] from `quicksyn_session.vszh5`, included as vector graphics. Commanded values are drawn as black diamonds.

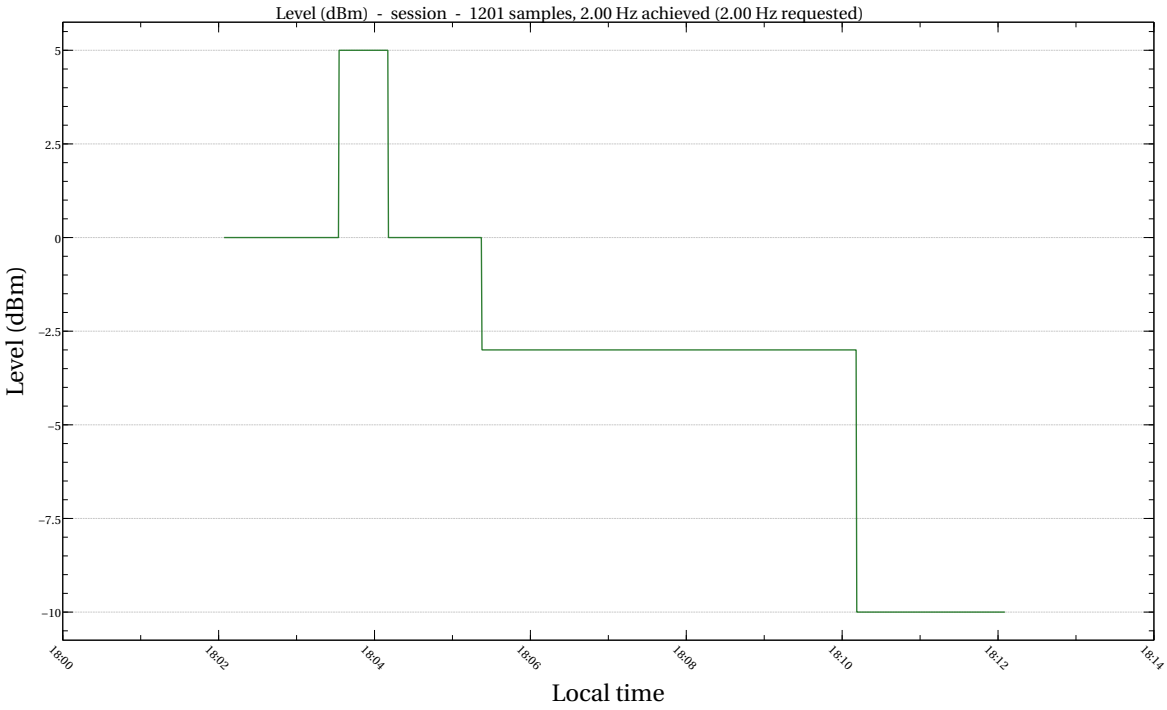
6.1 Overview (all quantities)



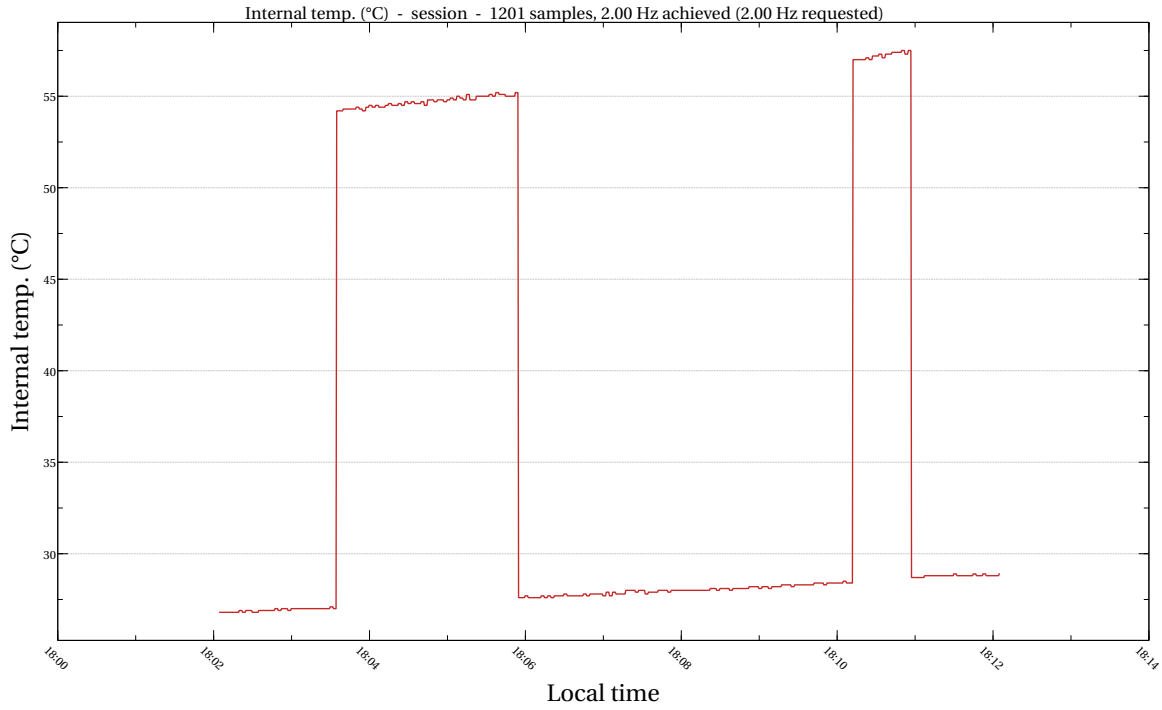
6.2 Carrier frequency



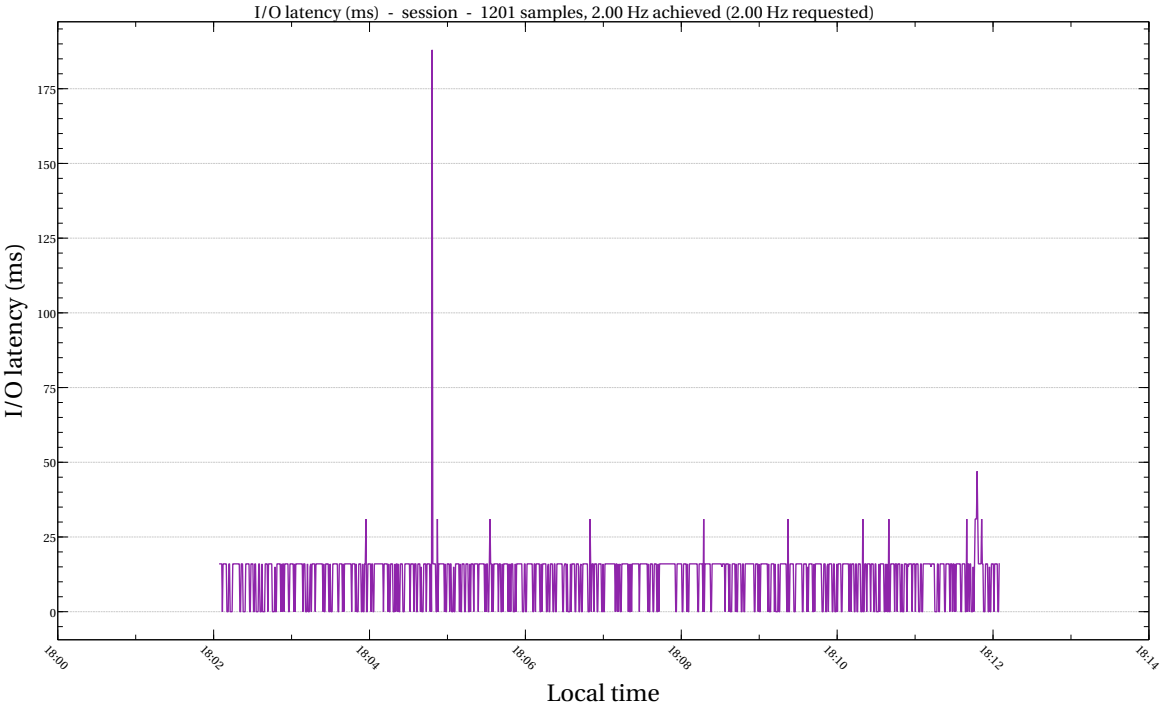
6.3 Output level



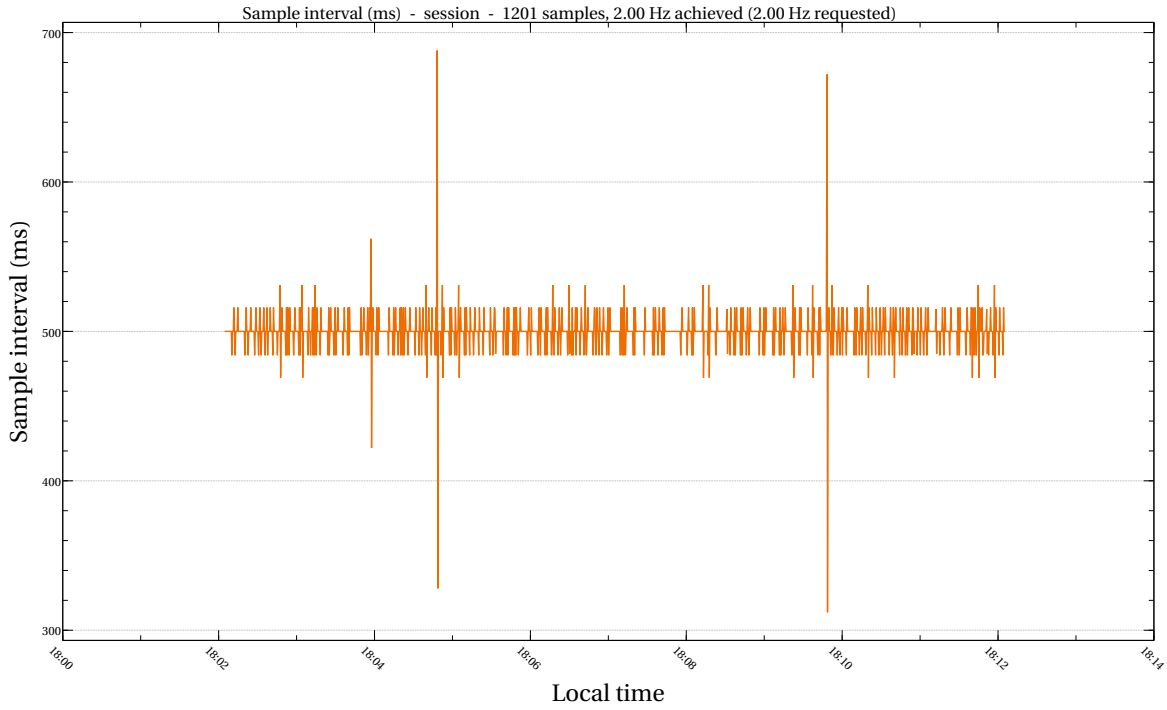
6.4 Internal temperature



6.5 Serial I/O latency



6.6 Sample interval



7 Files in the session folder

- `quicksyn_session.pdf`
- `quicksyn_session.vszh5`
- `quicksyn_session_p00_page_overview.png`
- `reduction_summary.json`
- `reduction_summary.md`
- `session.log`
- `session_metadata.json`
- `state_events.csv`
- `telemetry.csv`
- `telemetry_chunk_0000.pkl.gz`
- `telemetry_reduced.npz`
- `transactions.csv`
- `quicksyn_session_report.tex` and `quicksyn_session_report.pdf` (this report)

8 Provenance

Statistics computed with the numpy backend in 13.6 ms by quicksyn_veusz_export_v0_0_9.py 0.0.9. The same numbers are in `reduction_summary.json` and `reduction_summary.md`.

References

[1] National Instruments, "QuickSyn Microwave Synthesizers: Communications Specifications," Doc. No. 5580510-01, Santa Clara, CA, USA, 2014.

[2] National Instruments, "QuickSyn Microwave Frequency Synthesizer: User Guide," Doc. No. 5580513-01, Rev. F.

[3] National Instruments, "QuickSyn Frequency Synthesizers," product brochure and datasheet, Oct. 2017.

[4] *IEEE Standard Definitions of Physical Quantities for Fundamental Frequency and Time Metrology—Random Instabilities*, IEEE Std 1139-2008, 2008. [Online]. Available: `https://ieeexplore.ieee.org/document/4797525`

[5] Veusz Developers, "Veusz—a scientific plotting package." [Online]. Available: `https://veusz.github.io/`